

Communication-avoiding solvers for the FESOM2 sea-surface-height system

Interim report, 6 August 2026 — FESOM2 C++/Kokkos port, Levante (A100 GPUs, 128-core CPU nodes)

What was done. The barotropic sea-surface-height (SSH) solve is the model’s synchronisation bottleneck: preconditioned CG performs two global reductions and two halo exchanges per iteration. We implemented four alternatives behind a runtime switch — Chronopoulos–Gear CG (`cg2`), pipelined CG (`pipecg`), PIPECG-OATI (`oati`) and Chebyshev P-CSI (`pcsi`) — leaving the certified baseline bit-identical when the switch is unset. Two defects had to be fixed first. The preconditioner inherited from FESOM’s `solver.F90` is not symmetric (defect ratio 0.62–0.74 over three meshes), which invalidates the recurrence the CG–CG variants use in place of an explicit inner product: the step length came out wrong by 21.8% and most solves diverged. Symmetrising it also reduces the *existing* solver’s iteration count by about 7%. Our own stall guard was separately aborting converging solves, its threshold of 20 non-improving iterations sitting below the 21-iteration plateau baseline CG itself exhibits on fArc, so every previously reported fArc failure was a false positive.

Preliminary results. Each row is a same-allocation A/B against baseline CG, minimum of two repetitions, 300 steps, zero fallback firings. d_{SSH} is the solver’s own speed-up, d_{tot} the whole-step speed-up; the latter reproduces to 0.4 pp under a reversed leg order.

	configuration	vert./core	SSH %	best d_{SSH}	d_{tot}
CPU	CORE2, 512 ranks	248	9.9	−60.3 (<code>pcsi</code>)	−6.3
	CORE2, 864 ranks	146	18.9	−58.6 (<code>pcsi</code>)	−13.1
	fArc, 2048 ranks	312	22.8	−45.6 (<code>oati</code>)	−13.3
	DARS, 8192 ranks	385	6.4	−20.5 (<code>pcsi</code>)	−4.0
GPU	CORE2, 4 nodes	—	27.4	−18.3 (<code>oati</code>)	−5.2
	CORE2, 16 nodes	—	30.5	−20.5 (<code>oati</code>)	−6.7
	DARS, 8 nodes	—	5.7	none — all variants lose	

The SSH share of the step, not the mesh size or the per-core workload, predicts the payoff. It rises as a mesh is spread thinner — on CORE2 from 2.5% at 128 ranks to 18.9% at 864 — so the solvers pay most where the model stops scaling, moving CORE2’s parallel-efficiency knee out by a factor of 1.26–1.29. On GPU the share is 27–31% at ordinary node counts, because the device makes local work fast while the solve stays latency-bound. Where the share is small the variants lose, trading reductions for extra vector passes: a poor bargain when the solve is 6% of the step, as on DARS. `oati` is the better general choice and `pcsi` a per-mesh specialist, since `oati` tracks baseline CG’s iteration count everywhere while `pcsi` converges faster only where its Chebyshev interval is accurate — an eigenvalue estimate that was silently wrong on fArc.

Status and caveats. The fArc rows need the stall guard widened past its shipped default, and the value to ship has not been chosen. The GPU rows predate that fix; fArc GPU is withdrawn pending re-measurement, and NG5 has no valid CPU row, its tested rungs lying outside the operating range and its baseline leg failing at 4096 ranks and above. DARS is measured at 300–500 vertices per core on newly generated partitions, which form their own scaling curve. All numbers come from one machine and a 300-step window, and the baseline solver remains the shipped default.